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Artificial Intelligence

Master's Thesis

IMPROVING ROMANIAN ASR FOR SPECIFIC DOMAINS WITH TTS-BASED DATA AUGMENTATION

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Abstract

This thesis explores the use of text to-speech synthesis (TTS) as a data augmentation method to improve the performance of Automatic Speech Recognition (ASR) systems for low-resource languages in specialized domains. This study presents a novel dataset of Romanian audio-transcript pairs sourced from astrology-related YouTube videos, primarily designed for ASR but also applicable to other speech-related tasks such as language identification. We generate synthetic speech from text prompts automatically extracted from the horoscope section of ProTV, a major Romanian media TV channel. The text is converted to audio using TTS systems developed by Google and Microsoft. This synthetic data is then combined with the training data from our newly introduced dataset to fine-tune the Whisper Small model, which was pre-trained on large-scale multilingual data. Experimental results demonstrate improved performance on the test set of the new dataset as well as on two other Romanian horoscope-related datasets.

Rezumat

Lucrarea de față abordează utilizarea sistemelor de sinteză a textului în vorbire (TTS) ca metodă de augmetare a datelor pentru a îmbunătăți performanța sistemelor de recunoaștere automată a vorbirii (ASR) pentru limbi cu resurse limitate, în domenii specializate. Studiul prezintă un nou set de date format din perechi audio-transcriere în limba română, obținute din videoclipuri postate pe platforma YouTube pe subiecte legate de astrologie, conceput în principal pentru ASR, dar aplicabil și altor sarcini legate de vorbire, cum ar fi identificarea limbii. Generăm data sintentice de vorbire pe baza unor texte extrase automat din secțiunea horoscop a ProTv, una dintre cele mai importante televiziuni din România. Textele sunt convertite în fișiere audio utilizând sistemele TTS dezvoltate de Google și Microsoft. Aceste date sintetice sunt combinate cu datele de antrenare din noul set de date introdus pentru a realiza fine-tuning-ul variantei Small a modelelor Whisper, pre-antrenate pe un volum mare de date multilingve. Rezultatele experimentale demonstrează o îmbunătățire a performanței atât pe setul de test al noului dataset, cât și pe alte două seturi de date similare pentru limba română, legate de horoscop.

Contents

Chapter 1

Introduction

The **A**utomated **S**peech **R**ecognition (ASR) [10] task consists in the automated process of decoding and transcribing oral speech. It is fundamentally different from *speech understanding* - which concerns the meaning of speech, rather than its transcription - and *voice recognition* - which focuses on recognizing which person is speaking.

ASR systems have considerably advanced [23] since the introduction of the Whisper architecture - models built onto it achieved promising results for speech understanding and processing for an extended range of languages. However, results for low-resource languages are still considerably lower than the ones achieved for their counterparts, especially for Romanian, as observed in the results achieved on the CommonVoice [1] dataset, displayed in Table 1.1.

Table 1.1: **WER (%) on CommonVoice9.** *The table displays the WER values for the Whisper model variants for various languages from the CommonVoice dataset.*

Model	English	Romanian	Russian	Italian	French
Whisper tiny	28.8	68.2	40.6	49.6	49.7
Whisper base	21.9	55.9	28.8	36.1	37.3
Whisper small	11.2	33.2	15.0	16.0	22.7
Whisper medium	9.4	21.5	9.3	9.4	14.7
Whisper large	8.6	19.8	8.2	8.1	13.1
Whisper large-v2	6.4	15.8	7.1	7.1	11.7

1.1 Related work

Previous research has demonstrated [5] that augmenting accented speech datasets with synthetic data generated via Text-to-Speech (TTS) synthesis, combined with fine-tuning of Wav2Vec 2.0 [2] models, leads to significant improvements in Word Error Rate (WER) for speech recognition tasks.

The largest variant of the Whisper models [15] has outperformed its Wav2Vec-based counterpart across multiple datasets, including those containing Romanian, such as Common Voice [1] and FLEURS [3].

Moreover, fine-tuning Whisper models with additional data has shown significant improvements in WER for low-resource languages like Amharic [6]. This improvement particularly highlights the effectiveness of language-specific normalization techniques, such as homophone normalization, in enhancing ASR performance.

1.2 Objectives

The current study seeks to extend the findings presented in Section 1.1 by investigating potential improvements in WER results for speech recognition, applied to domain-specific Romanian data within the specialized field of astrology. Accordingly, the preliminary objectives are outlined as follows:

1. *The development of a specialized-domain (Astrology) Romanian dataset for ASR*: The dataset consists of audio-transcript pairs extracted from publicly available videos posted on the Youtube platform. The transcripts have been curated through a semi-automated annotation process which implied the manual annotation based on the automatically-generated subtitles of the video provided by Youtube. Afterwards, the annotations have been checked and adjusted upon further comparison with transcripts computed with the Large variant of the Whisper-timestamped model.

2. *Fine-tune Whisper-based models on the newly developed dataset and the aug-*

mented variations containing synthetic data: The Small variant of the Whisper model was selected for the experiments due to limited computational resources. The experimental setup was carefully designed to assess the impact of synthetic data augmentation on Romanian ASR performance within the specialized domain of astrology, including augmentations using diverse subsets of synthetic data, varying in both duration and textual content.

Chapter 2

Dataset

2.1 Real data

The dataset, which is publicly available[1], is designed for the ASR task and includes audio files, along with their corresponding text transcripts. With audio and text data exclusively in Romanian, the dataset aims to contribute to the continuous efforts to enhance language support for low-resource languages - especially when it comes to specialized domains such as horoscope. Dataset-specific details such as the total number of samples, speakers, duration per split are available in Table 2.1.

Table 2.1: **Statistics for real data corpus.** *The table describes the total number of samples (pairs of audio-transcript), speakers, duration for each split of the real-data corpus.*

Split	Samples	Speakers	Duration	Audio sample duration		
				Min	Avg	Max
Train	413	3	2:54:42	0:00:03	0:00:25	0:00:30
Validation	90	2	0:36:37	0:00:02	0:00:24	0:00:30
Test	92	4	0:36:44	0:00:04	0:00:23	0:00:30

The training subset of the dataset contains audio recordings from three different

[1] https://huggingface.co/datasets/IoanaLiviaPopescu/RealVoiceHoroscope Real data dataset

speakers: Acvaria, Andreea Dincă and Remus Ionescu. For Acvaria, videos from the *Horoscopul săptămânii* (weekly horoscope) series were selected. For Andreea Dincă, samples are drawn from both the *Horoscop de weekend* (weekend horoscope) and *Horoscopul săptămânii* (weekly horoscope) series. Meanwhile, the selected material for Remus Ionescu consists of videos from the television series *AstroBani* hosted on the *Playtech Știri* news channel, which focuses on weekly horoscope from financial perspectives. Associated statistics with the train split of the dataset are displayed in Table 2.2.

Table 2.2: **Statistics for speakers - train split.** *The table displays gender, total number of samples, total duration with associated audio files and minimum, maximum and average duration for audio samples for the speakers in the train dataset.*

textbfSpeaker	**Gender**	**Samples**	**Duration**	**Audio sample duration**		
				Min	**Avg**	**Max**
Acvaria	Female	157	1:04:54	0:00:08	0:00:24	0:00:30
Andreea Dincă (Astrostyle)	Female	158	1:10:50	0:00:03	0:00:26	0:00:30
Remus Ionescu	Male	98	0:38:53	0:00:05	0:00:23	0:00:29

The validation subset comprises audio material from the *Horoscopul săptămânii* series of two speakers: Thomas Rob, which presents the series on the *WOWnews* Youtube channel, and Alexandra Coman - additional samples have been collected from the *Astroshorts* edition as well. Associated statistics with the validation split of the dataset can be viewed in Table 2.3.

Table 2.3: **Statistics for speakers - validation split.** *The table displays gender, total number of samples, total duration with associated audio files and minimum, maximum and average duration for audio samples for the speakers in the validation dataset.*

Speaker	**Gender**	**Samples**	**Duration**	**Audio sample duration**		
				Min	**Avg**	**Max**
Alexandra Coman	Female	45	0:17:49	0:00:02	0:00:24	0:00:30
Thomas Rob	Male	45	0:18:48	0:00:12	0:00:25	0:00:30

The test subset includes audio data from four speakers: two females, Camelia Pătrășcanu and Maria Sârbu, and two males, Dan Ciubotaru and Valeriu Panoiu. The audio samples from Camelia, Maria, and Valeriu cover topics similar to those found in the training and validation sets. Specifically, Valeriu's recordings are from *Cronicile astrale* (Weekly Horoscope), Maria's from *Horoscop săptămânal*, and Camelia's from *Horoscop lunar* (Monthly Horoscope). Due to the limited availability of Romanian horoscope data, especially for male speakers, the samples from speaker Dan Ciubotaru focus on astrology-related topics with a stronger emphasis on historical perspectives: *România 2025-2029 - perspectivă astrologică* (*Romania 2025–2029: An Astrological Perspective*) and *Anul 2025 din perspectivă astrologică* (*The Year 2025 from an Astrological Perspective*). Dataset-specific statistics for the test split are presented in Table 2.4.

Table 2.4: **Statistics for speakers - test split.** *The table displays gender, total number of samples, total duration with associated audio files and minimum, maximum and average duration for audio samples for the speakers in the test dataset.*

Speaker	Gender	Samples	Duration	Audio sample duration		
				Min	Avg	Max
Camelia	Female	28	0:10:33	0:00:04	0:00:22	0:00:29
Maria	Female	23	0:08:17	0:00:09	0:00:21	0:00:29
Dan	Male	21	0:08:46	0:00:10	0:00:25	0:00:29
Valeriu	Male	20	0:09:07	0:00:18	0:00:27	0:00:30

Data collection

The audio data has been collected from public videos posted on the Youtube platform from various astrologists, containing a mix of profesional and home-made videos. The options are scarce, especially in comparison with the amount of available videos in English, perhaps due to speakers choosing English over their native language in an effort to reach a larger audience. Also notable is the predominance of female Romanian astrologists, which also reflects in the gender distribution of the final dataset sample count and total duration.

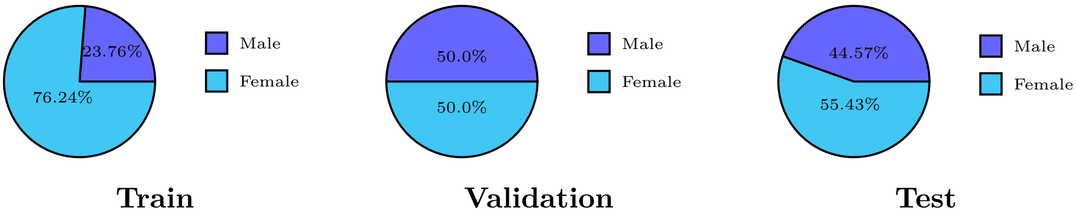

Figure 2.1: **Gender distribution for the real-data corpus in relation to audio sample count.**

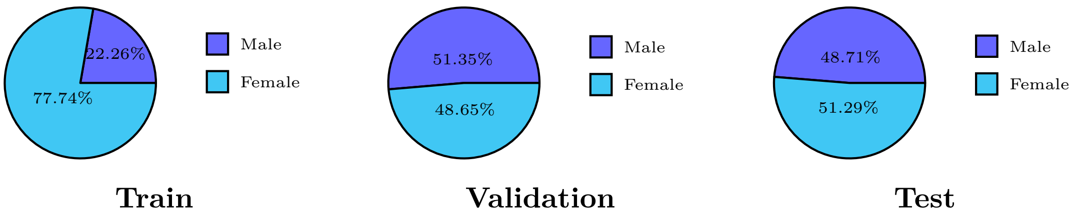

Figure 2.2: **Gender distribution for the real-data corpus in relation to total duration of audio samples.**

Audio data from Youtube videos has been downloaded using one of the most popular dedicated tools - *yt-dlp* [4] in the WAV (**Wav**eform Audio File Format) format. Relevant utilized command-line options for preprocessing are presented below:

- Text data

```
 --write-auto-sub
--sub-lang", "ro"
--convert-subs", "srt"

```

The above command-line options manage the generation of Romanian subtitles automatically in the .srt file format.

```
--use-postprocessor", "srt_fix"

```

The initial .srt file contained duplicate lines in the subtitles. This option has been introduced to mitigate this issue, generating another file with the fixed subtitles.

- Audio data

```
    --extract-audio"
    --audio-format", "wav"
    --audio-quality", "0"
    --postprocessor-args", "ffmpeg:-ar 16000"

```

  The above command-line options configure the extraction of audio data in the WAV format with a 16000Hz sample rate.

Data preprocessing

The next step implied dividing the audio files in chunks of maximum 30 seconds - with respect to Whisper's receptive field. To avoid cutting words, a **V**oice **A**ctivity **D**etection (VAD) tool can be leveraged to obtain timestamps that identify the periods during which the speaker is talking. The Silero VAD [19] model has been chosen, the family of [18] Silero Models being previously used in related articles exploring topics related to fine-tuning Whisper models on low-resource languages, such as [21] (Swiss German).

Upon obtaining the VAD timestamps for a video, we iterate through them, merging adjacent chunks into larger ones, with respect to the upper threshold of 30 seconds. The timestamps corresponding to chunks larger than 30 seconds were discarded.

In order to obtain the desired audio-transcript pairs, .txt files were created by merging the subtitles of whose ranges intersected with the [start, end] timestamps of each chunk. As both the silero-VAD model and Youtube auto-generate subtitle algorithm do not have perfect accuracy, the created .txt files often contained overlapping words with the previous or next chunk.

Data annotation

- First iteration

During the first iteration, a single native speaker annotated the dataset based on the initial support provided by the associated .txt files previously mentioned, with respect to casing, punctuation and diacritics. Audio samples that were considered of poor quality (ambigous pronounciation, stutter, noise) have been discarded.

- Second iteration

  In an effort to overcome the limitation of a single annotator, a second iteration has been conducted. On the previously selected audio samples, a Whisper-timestamped [11] model has been leveraged in order to extract identified words and their associated timestamps which offer finer granularity compared to the initial VAD timestamps. During this iteration, certain audio samples were further trimmed to eliminate noise that , while undetectable by the annotator, has been identified by the model. Furthermore, errors in the previous annotations have been identified - including typos and misinterpreted words.

  The statistics regarding the applied changing during the second iteration of the annotation process can be viewed in Table 2.5.

Table 2.5: **Statistics for second iteration real data dataset annotation changes.** *Trimmed samples involved adjusting the audio in length in order to remove noise. Corrected samples involved fixing errors in the previously computed text annotations. Discarded samples have been eliminated from the dataset due to low audio quality - ambiguous pronounciation, stutter, noise.*

Split	Trimmed samples	Corrected samples	Discarded samples
Train	22 (5.65%)	23 (5.91%)	13 (3.34%)
Validation	4 (4.12%)	7 (7.69%)	1 (1.09%)
Test	2 (2.17%)	5 (5.43%)	0 (0%)

2.2 Synthetic data

Data collection

Text data

The text data used to generate the synthetic datasets through TTS (**T**ext **T**o **S**peech) was sourced from the public horoscope section of the official ProTV website [17]. ProTV is a well-known television platform in Romania, ranking among the top in audience viewership. The horoscope section features daily articles that provide a brief set of sentences for each zodiac sign, offering daily guidance and predictions.

The text content of the horoscope articles from pages 1 to 50 [2] was automatically extracted via web scraping with the help of BeautifulSoup [16], one of the most widely used Python libraries for parsing HTML content. Initially, article links were retrieved by identifying and targeting associated HTML element tags - due to frequent updates on the website observed in the following days, specific tag references will be omitted, as they are no longer applicable. Once the links were collected, the content of each article was accessed through the *article–text* class tag, and the text from the paragraph elements within it, as displayed below.

```
<div class="article--text">
    <p>
        "Extracted text sample."
    </p>
</div>
```

To avoid a wide range of short audio samples, paragraphs that were empty or contained fewer than 70 characters were discarded. This procedure yielded a total of 12036 text samples, comprising 2335901 characters. An initial audio dataset has been generated using the *ro-RO-Standard-B* voice (Google TTS). The text entries for which the generated audios had a duration larger than 30 seconds were discarded, resulting in 11692 remaining samples - which have then been shuffled.

[2]As of May 30, 2025, 08:00 AM (UTC)

Data preprocessing

However, the extracted text data presented a few issues that would greatly impact the results of TTS: numbers in their digit format (with which we discovered the chosen TTS solutions struggle), formatting errors (such as missing spaces, leading to concatenated words).

In order to mitigate the potential negative impact of these errors on the quality of the synthetic data, a solution leveraging GPT-4 [14] capabilities has been employed. Thus, the following prompt has been utilized for *gpt-4.1-nano* model, the cost-effective solution recommended for simpler tasks.

```
Ești un asistent care corectează textele în limba română.
Corectezi greșelile gramaticale, de ortografie, de punctuație
și de majuscule.
Corectezi acordurile greșite (inclusiv între numerale și
substantive).
În plus, transformi toate numerele scrise ca cifre (ex: 2, 15, 123)
în forma lor scrisă cu litere, păstrând sensul frazei.
Răspunde doar cu textul corectat, fără explicații.
```

The prompt has been validating upon analyzing the results on various dummy examples, such as the one displayed in Table 2.6. An example of an actual corrected text sample from the final dataset can be viewed in Table 2.7.

Table 2.6: **Example of corrected dummy text with GPT-4.** *A dummy example has been created in order to test the quality of the utilized prompt and GPT-4 capabilities. The text has been correctly adjusted.*

Version	**Text**
Initial	Am cumparat 2 mere si 2 morcovi. Nu stiu daca in 2026 vor creste preturile. Oare chiar te crede !
Corrected	Am cumpărat două mere și doi morcovi. Nu știu dacă în două mii douăzeci și șase vor crește prețurile. Oare chiar te crede ?

Table 2.7: **Example of corrected text sample from corpus with GPT-4.** *The example displays an actual text sample for the corpus utilized for generating synthetic audio data. The text has been correctly adjusted.*

Version	**Text**
Initial	În săptămâna analizată, nativii Scorpionsunt interesați de rezolvarea unui context privind o relație de muncă, cu șefii, colegii sau personală, privind activitățile, rutina zilnică și sănătatea.
Corrected	În săptămâna analizată, nativii Scorpion sunt interesați de rezolvarea unui context privind o relație de muncă, cu șefii, colegii sau personală, privind activitățile, rutina zilnică și sănătatea.

Audio data

The audio data has been generated with the help of the TTS solutions provided by some of the top companies with such offerings (Google, Microsoft). The utilized voices, alongside key-details, can be viewed in Table 2.8.

Table 2.8: **Selected synthetic voices from TTS solutions offered by providers.** *The voices listed below have been utilized to generate the synthetic audio data used for augmentation in the presented experiments.*

Voice	Provider	Gender
ro-RO-Wavenet-B	Google	Female
ro-RO-Standard-B	Google	Female
ro-RO-EmilNeural	Azure	Male

The speech samples have been obtained automatically, following the official guides regarding the use of the API with Python for extraction in the WAV format. All of the audio entries are automatically resampled at 16000Hz during dataset loading, via the dataset script.

In terms of the models utilized for TTS:

- **Azure** : The official documentation does not disclose the model architecture; only mentions the use of ”deep neural networks” [20], offering ro-RO-AlinaNeural (Female) and ro-RO-EmilNeural (Male) for Romanian.

- **Google** : The ro-RO-Wavenet-B voice is based on WaveNet [13], developed by Google DeepMind and introduced in 2016 - the first version showcasing state-of-the-art results for English and Mandarin TTS. It is presented as a premium option, generating audio with a high resemblance to human speech. On the other hand, the standard voice options offered by Google, such as ro-RO-Standard-B, are based on traditional TTS methods - some use a variation of the parametric text-to-speech technology [7].

Chapter 3

Approach

3.1 Model

Whisper [15] introduces a suite of models built onto a newly introduced architecture in the context of ASR systems. This new approach involves chunking the audio input in 30 seconds, to be later converted into a log-Mel spectrogram. The spectogram is then passed to a Transformer-based component to extract high-level acoustic features, which are subsequently encoded into contextual representations and decoded into text sequences by a Transformer decoder, enabling robust, multilingual speech recognition and translation.

The models have been trained on a vast amount of real data, including 117,000 hours of speech across 99 languages (Romanian among them), in addition to 563,000 hours of English speech. The models are available in six variants which differ in size, with parameter counts ranging from 39M to 1550M. Larger models offer better transcriptions at the cost of a higher demand in terms of computational resources.

For the current experiments, the Small variant of the Whisper models was selected as a compromise between transcription performance and computational efficiency.

3.2 Fine-tuning process

The fine-tuning process was carried out following the guidelines from the Whisper Fine-Tuning Event [9], with minor modifications such as adopting epoch-based training, evaluation, and logging. These adjustments were made due to the small dataset size and to ensure a stable configuration across different experiments. The values chosen for training hyperparamters are displayed in Table 3.1.

The hyperparameter values have been adjusted to model size (whisper-small) and available computing resources (NVIDIA TESLA P100 GPUs - 16 GB - via Kaggle). As advised for memory-restricted settings, the Adam 8bit optimizer was chosen, allowing the use of larger batch sizes.

These settings are similar to the ones used in other publications regarding fine-tuning Whisper models on low-resources languages, such as Swiss German [21] or Catalan, Galician, Basque [23].

Table 3.1: **Training Hyperparameters.**

Parameter	**Value**
Optimizer	Adam_8Bit
Learning rate	$1e^{-5}$
Train Batch Size	16
Gradient Acc Steps	2
Eval Batch Size	8
Warmup ration	0.1
Number of epochs	5
Optimizer Specific Parameters	
β_1	0.9
β_2	0.999
ϵ	1.0×10^{-8}

3.3 Evaluation

Text standardization

In the previously mentioned tutorial [9] reference for the Whisper fine-tuning process, the recommended setting for the *do_normalize* flag is *True* which applies text standardization before computing WER. The *do_lower_case* and *do_remove_punctuation* flags have been set to *False*, as indicated, thus our fine-tuned models will output punctuation and casing, however, errors regarding these aspects will not be penalized. An example of applied text standardization can be observed in Table 3.2.

The text standardization method is the one described in the original Whisper [15] paper - a general approach applied for non-English text, regardless of the specific language. This is a limitation and area of future work identified by the authors - especially in comparison to the approach designed for English.

Table 3.2: **Example of text standardization results.**

Initial	Horoscop: Gemenii vor avea parte de o surpriză.
Normalized	horoscop gemenii vor avea parte de o surpriză

Evaluation metrics

WER [22] is a common metric for the ASR task, esentially measuring the *"cost of restoring the output word sequence to the original input sequence"* [12], according to the following formula:

$$\text{WER} = \frac{S + D + I}{N}$$

S = the total amount of substitutions

D = the total amount of deletions

I = the total amount of insertions

N = the total amount of words in the reference transcript

Concerns related to WER's capacity to accurately capture the actual level of human language understanding have been raised - among the issues is that it treats all error types equally, which may not reflect the actual impact on comprehension.

The metric (implementation [8]) has been chosen for simplicity and ease of potential future performance comparison on other datasets, as some of the benchmark datasets for ASR (such as CommonVoice [1] - with over 29 languages) report results on the WER metric. Table 3.3 provides a sample calculation of WER.

Table 3.3: **Example of WER computation results.**

Source	**Normalized text**	**WER (%)**
Ground-Truth	horoscop gemenii vor avea parte de o surpriză	-
Prediction	horoscop gemenii poate vor avea parte o surpriză	0.25

Explanation We have $N = 8$ total words in the normalized reference text (ground-truth). The word *poate* in the normalized prediction text does not appear in reference, thus is an insertion $I = 1$ and the word *de* from reference is missing $\rightarrow D = 1$. Upon introducing these values in the formula we obtain:

$$\text{WER} = \frac{0+1+1}{8} = \frac{2}{8} = \frac{1}{4} = 0.25$$

3.4 Experiments

Fine-tuning on real data versus fine-tuning on synthetic data

To accurately compare the performance of real versus synthetic data, audio data was generated based on the text entries of the real-data corpus presented in Section 2.1 train split using the same TTS tools to be used for augmentation, described in Table 2.8.

The *synth-data-v0* model has been fine-tuned on the SynthVoiceDataset-v0[1] dataset which has been obtained by generating audio data with the *ro-RO-Wavenet-B* voice for

[1] https://huggingface.co/datasets/IoanaLiviaPopescu/SyntheticVoiceHoroscope-v0

text entries corresponding to the speaker Acvaria, *ro-RO-Standard-B* for text entries corresponding to the speaker Andreea Dincă and *ro-RO-EmilNeural* for text entries corresponding to the speaker Remus Ionescu. The synthetic voices were selected according to the gender of the original speaker.

Similarly to *synth-data-v0*, the *synth-data-v1* model has been fine-tuned on the SynthVoiceDataset-v1[2] dataset which has been obtained by generating audio data with the *ro-RO-Standard-B* voice for text entries corresponding to the speaker Acvaria, *ro-RO-Wavenet-B* for text entries corresponding to the speaker Andreea Dincă and *ro-RO-EmilNeural* for text entries corresponding to the speaker Remus Ionescu. Analogously, the synthetic voices were selected according to the gender of the original speaker.

Data augmentation

Data augmentation implies augmenting the initial dataset - containing real data, previously presented in Section 2.1 - with an amount of synthetic data - obtained and preprocessed as presented in Section 2.2. We aim to study if and how performance improves - if yes, at what rate and how consistent and if not, what errors are being made and their potential causes.

The name of the augmented datasets created is in the format:

`{number_of_samples}_{split_id}_{voice_option}`

- *number_of_samples* : the number of text entries for which audio was generated and added alongside the real training data (e.g. 400 if synthetic audio data based on 400 text samples has been used to augment the initial real-data corpus)

- *split_id* : based on the initial text samples curated from the ProTv website, multiple subsets of different sizes have been created - we will refer to them as splits (e.g. split 400_1 refers to the first subset of size 400 out of the selected text samples; 400_2 is the second subset created and has no samples in common with the other subsets with the same size, such as 400_1).

[2] https://huggingface.co/datasets/IoanaLiviaPopescu/SyntheticVoiceHoroscope-v1

- *voice_option* : refers to the synthetic voice used to generate the audio data based on the selected text data (e.g. 401_1_Standard-B refers to the dataset which has been augmented with audio data generated with the ro-RO-Standard-B voice from Google TTS based on the subset of the first 400 samples of text data selected out of the total scraped text data from ProTv; for mixed datasets such as 801_1_Mixed-St-Wav, *St* stands for ro-RO-Standard-B and means the first half of the 800 text samples has been generated with it and the rest with the second voice, as *Wav* stands for ro-RO-Wavenet-B (Google TTS)).

Table 3.4 summarizes the synthetic datasets generated for augmenting the real-data corpus and the total duration each adds.

Table 3.4: **Corpora created for synthetic data augmentation.** *The column Augmentation data refers to the selected audio-transcript pairs to be added to the real-data corpus for fine-tuning the model. The other column displays the total duration that will be added, not counting the duration of the real-data corpus.*

Augmentation data	Augmented data duration
400-1-Standard-B, 400-1-Wavenet-B	1:18:00
400-1-EmilNeural	1:15:35
400-2-Standard-B. 400-2-Wavenet-B	1:13:06
400-2-EmilNeural	1:11:13
800-1-Standard-B, 800-1-Wavenet-B	2:31:06
800-1-EmilNeural	2:27:48
1200-1-Standard-B, 1200-1-Wavenet-B	3:46:28
1200-1-EmilNeural	3:41:23
1600-1-Standard-B, 1600-1-Wavenet-B	5:05:48
1600-1-EmilNeural	4:56:26
2000-1-Standard-B, 2000-1-Wavenet-B	6:21:49
2000-1-EmilNeural	6:10:04

Observation Subset *800_1* is formed out of the corresponding text and audio samples of *400_1* and *400_2*. Following the same pattern, subset *1200_1* contains samples from *800_1*, augmented with another 400 samples (different from the ones in *400_1*, *400_2*); the same logic applies to subset *1600_1* in relation to *1200_1*.

3.5 Results

Benchmark

Whisper models

To establish a baseline for future fine-tuning experiments, we first evaluate the Whisper models on the referenced test set. In addition, we assess their performance on the test split of two other Romanian ASR datasets, which feature samples from YouTube videos of astrologers Neti Sandu[3] and Urania[4]. The baseline results are displayed in 3.5.

Table 3.5: **WER (%) results - whisper models.** *WER results of Whisper model variants computed on the test splits of the Urania, Neti and RealVoiceDataset (real-data corpus).*

Model	Urania	Neti	RealVoiceDataset				
			Camelia	Maria	Dan	Valeriu	Overall
whisper-tiny	83.43	77.73	58.81	57.55	77.08	71.65	67.19
whisper-base	58.37	61.50	38.43	44.17	67.65	44.86	48.22
whisper-small	25.99	35.33	17.06	27.07	43.98	24.22	27.26
whisper-medium	12.62	15.60	6.51	17.26	26.62	9.72	14.48
whisper-large	8.37	19.44	5.00	14.00	24.34	8.59	12.44

Fine-tuning on real data versus fine-tuning on synthetic data

Next, we compare the performance of real data versus synthetic data by fine-tuning the same model (whisper_small) on the same text data, but different voices (real and

[3] https://huggingface.co/datasets/iulik-pisik/horoscop_neti
[4] https://huggingface.co/datasets/iulik-pisik/horoscop_urania

two various of speaker to voice associations for the synthetic data, with respect to the gender of the speaker).

Table 3.6: **WER (%) results - real vs synthetic data (whisper-small).** *WER results of the Small Whisper model variant fine-tuned on the real-data corpus versus on the SynthVoiceDataset-v0 and SynthVoiceDataset-v1, previously introduced in Section 3.4*

Split	**Urania**	**Neti**	**RealVoiceDataset**				
			Camelia	**Maria**	**Dan**	**Valeriu**	**Overall**
benchmark	25.99	35.33	17.06	27.07	43.98	24.22	27.26
real-data	16.89	18.39	12.65	13.54	33.49	16.26	18.41
synth-data-v0	17.15	21.70	12.61	15.67	36.37	15.69	19.47
synth-data-v1	17.12	21.50	12.66	15.67	36.15	15.69	19.43

The table above displays how the model fine-tuned on real data outperforms both models fine-tune on synthetic audio data generated based on the real-data text transcripts. There are two exceptions - model *synth-data-v0* has a slightly better score (0.04%) for one of the speakers in RealVoiceDataset (real data corpora) and both synthetic models have a better result on the subset of the speaker Valeriu, with an error decrease of (0.57%). However, both models fine-tuned on synthetic data still show a considerable improvement compared to baseline which is promising for further experiments based on synthetic data augmentation.

Data augmentation

Subsequently, several datasets built by augmenting the initial real data corpus with synthetic data were created to properly study how performance is impacted by voice variability and synthetic dataset size.

The results obtained for the audio data generated with the *ro-RO-Standard-B* voice are displayed in Table 3.7. This approach achieves a maximum *2.33%* overall WER improvement for the test split of the real data corpus, *2.05%* for Neti and *1.22%* for Urania.

Table 3.7: **WER (%) results - ro-RO-Standard-B**. *WER results of the Small Whisper model variant fine-tuned on the real-data corpus augmented with audio data generated with the ro-RO-Standard-B voice from Google.*

Dataset	Urania	Neti	RealVoiceDataset				
			Camelia	Maria	Dan	Valeriu	Overall
real-data (benchmark)	16.89	18.39	12.65	13.54	33.49	16.26	18.41
400-1-Standard-B	16.17	17.29	12.79	12.34	31.82	15.03	17.51
400-2-Standard-B	16.69	17.26	12.34	11.88	32.57	15.23	17.47
800-1-Standard-B	16.20	17.43	11.57	11.62	32.27	14.63	16.97
1200-1-Standard-B	15.67	16.91	10.55	10.94	31.03	14.43	16.16
1600-1-Standard-B	21.48	16.34	10.34	10.32	31.36	14.72	16.08
2000-1-Standard-B	15.90	16.05	11.13	11.22	30.66	14.03	16.24

The results obtained for the audio data generated with the *ro-RO-Wavenet-B* voice are displayed in Table 3.8. This approach achieves a maximum *2.44%* overall WER improvement for the test split of the real data corpus, *2.08%* for Neti and *1.26%* for Urania.

Table 3.8: **WER (%) results - ro-RO-Wavenet-B**. *WER results of the Small Whisper model variant fine-tuned on the real-data corpus augmented with audio data generated with the ro-RO-Wavenet-B voice from Google.*

Dataset	Urania	Neti	RealVoiceDataset				
			Camelia	Maria	Dan	Valeriu	Overall
real-data (benchmark)	16.89	18.39	12.65	13.54	33.49	16.26	18.41
400-1-Wavenet-B	16.18	17.31	12.83	12.59	31.91	14.92	17.58
400-2-Wavenet-B	16.25	17.32	12.31	11.43	32.59	14.68	17.23
800-1-Wavenet-B	16.26	17.36	11.57	11.62	32.27	14.69	16.99
1200-1-Wavenet-B	15.63	16.76	10.65	11.50	31.11	14.35	16.34
1600-1-Wavenet-B	16.08	16.31	10.44	10.32	30.65	14.80	15.97
2000-1-Wavenet-B	16.10	15.70	10.99	11.12	31.70	13.44	16.28

The results obtained for the audio data generated with the *ro-RO-EmilNeural* voice are displayed in Table 3.9. This approach achieves a maximum *3.21*% overall WER improvement for the test split of the real data corpus, *2.96*% for Neti and *1.97*% for Urania.

Table 3.9: **WER (%) results - ro-RO-EmilNeural**. *WER results of the Small Whisper model variant fine-tuned on the real-data corpus augmented with audio data generated with the ro-RO-EmilNeural voice from Azure (Microsoft).*

Dataset	**Urania**	**Neti**	**RealVoiceDataset**				
			Camelia	**Maria**	**Dan**	**Valeriu**	**Overall**
real-data (benchmark)	16.89	18.39	12.65	13.54	33.49	16.26	18.41
400-1-EmilNeural	15.54	17.48	13.56	13.82	32.02	17.63	18.72
400-2-EmilNeural	16.46	16.59	12.02	12.93	31.33	14.68	17.23
800-1-EmilNeural	16.08	17.03	11.47	11.31	29.06	14.18	16.03
1200-1-EmilNeural	15.38	15.43	11.98	10.70	28.76	13.71	15.87
1600-1-EmilNeural	14.92	15.97	11.37	10.75	26.66	13.65	15.20
2000-1-EmilNeural	14.74	14.77	10.72	10.09	26.75	13.14	14.75

The results obtained for the audio data generated with multiple voices are displayed in Table 3.10. This approach achieves a maximum *3.66*% overall WER improvement for the test split of the real data corpus, *3.62*% for Neti and *2.15*% for Urania.

Table 3.10: **WER (%) results - Mixed**. *WER results of the Small Whisper model variant fine-tuned on the real-data corpus augmented with mixed audio data - if two voices are listed, the first half of the audio data was generated with the first voice, the other with the second. If there are three voices, the first third is generated with the first voice, second third with second voice and third with the last voice. St stands for ro-RO-Standard-B, Wav for ro-RO-Wavenet-B and Emil for ro-RO-EmilNeural.*

Dataset	Urania	Neti	RealVoiceDataset				
			Camelia	Maria	Dan	Valeriu	Overall
real-data (benchmark)	16.89	18.39	12.65	13.54	33.49	16.26	18.41
800-1-Mixed-St-Wav	16.11	17.40	11.62	11.62	32.19	14.63	16.97
1200-1-Mixed-St-Wav-Emil	15.61	16.10	11.04	10.16	32.33	14.45	16.24
1600-1-Mixed-St-Emil	15.62	16.15	11.08	10.16	32.33	14.67	16.48

Table 3.11: **Best WER (%) for each dataset**. *Best WER results per dataset out of all of the conducted experiments presented above.*

Dataset	Urania	Neti	RealVoiceDataset				
			Camelia	Maria	Dan	Valeriu	Overall
real-data (benchmark)	16.89	18.39	12.65	13.54	33.49	16.26	18.41
1600-1-EmilNeural	14.92	15.97	11.37	10.75	26.66	13.65	15.20
1600-1-Standard-B	21.48	16.34	10.34	10.32	31.36	14.72	16.08
2000-1-EmilNeural	14.74	14.77	10.72	10.09	26.75	13.14	14.75

3.6 Performance and error analysis

Among the main observations is that increasing the amount of synthetic data added - from 400 to 2000 with a step size of 400 audio samples (corresponding to approximately one to six additional hours added to the initial three hours of real data) - generally improves performance on the test datasets with as much as *3.66%* for the RealVoiceDataset, (*6.83%* for the audio samples collected for Dan Ciubotaru), respectively *2.15%* for Urania and *3.62%* for Neti.

Experiments have been conducted on samples generated with every voice previously presented in Table 2.8. While results for *ro-RO-Wavenet-B* and *ro-RO-Standard-B* are similar and seem to follow the same trends, the performance of *ro-RO-EmilNeural* is much higher, as the model trained with the largest amount of synthetic data generated with it achieved top results for the majority of test subsets, as shown in Table 3.11.

Moreover, augmenting with a synthetic dataset containing with samples generated with different voices shows promising results, as observed in Table 3.10. The fine-tuned model on the augmented dataset *1200-1-Mixed-St-Wav-Emil* (1200 audio samples, out of which subsets of 400 were generated with ro-RO-Standard-B, ro-RO-Wavenet-B, ro-RO-EmilNeural) outperforms the model fine-tuned on the *1600-1-Mixed-St-Emil* which contains a higher amount of synthetic data.

Augmenting with 400 to 1600 audio samples (one to five hours in duration), there seems to be a general increasing trend in performance for each of the voices. However, when further increasing the amount of synthetic data to 2000 samples (around six hours), WER results continue to improve for some of the datasets, however, there is a slight increase in error for the rest of the datasets. This phenomenon can be noticed in Table 3.8 and it seems to be influenced by the utilized synthetic voice, as for *ro-RO-EmilNeural*, results seem to continue to improve.

When analyzing the errors present in the normalized predictions, one of the aspects the models seem to struggle the most is numeric data. In the train, validation and test splits of the utilized datasets - including the synthetic data utilized for augmentation - numbers are present in their written form. The utilized text standardization method does not handle the conversion of written form in Arabic form or viceversa, thus penalizing errors such as *5* instead of *cinci*. As these differences do not play a role in semantics, an alternative text standardization adapted to Romanian could be further developed to better assess performance. Other errors regarding numbers involve spacing (e.g. *trei zeci* instead of *treizeci*) - the models fine-tuned on the larger variations of the augmented datasets proved to handle numbers better.

Identified types of errors and model-specific performance are highlighted in the examples presented in Appendix A. Moreover, the examples show how fine-tuning of data from specialized domains improves the transcription of specific terms (e.g. *Jupiter*). Furthermore, errors such as out-of-vocabulary words (e.g.*vastră* instead of *voastră*) are also present, thus incorporating a language model to address this issue could further advance performance.

Chapter 4

Conclusions and future work

In conclusion, the conducted experiments show that generally, augmenting datasets with synthetic data obtained through Text-to-Speech (TTS) yields improved WER results on specialized datasets for low resource languages - exemplified by the particular case of Romanian data on astrology-related topics. The quality of the transcriptions is influenced by a multitude of factors, how many and which voices were used, text content and normalization techniques.

In terms of future work, a novel text standardization method for Romanian may be introduced in order to assess WER results more accurately - in particular, high errors were generated by Whisper models either mispelling the written form of numbers or utilizing the Arabic notation. In Romanian, numbers cannot simply be converted to their written form without taking into account the context, as it may differ from the form with article. Moreover, transformations applied to English such as contraction expansion or conversion of informal or regional spellings into standardized ones may also improve accuracy if adapted appropriately for Romanian.

Future experiments involving a larger amount of synthetic data should be conducted to evaluate the potential for continuous improvements. Additional synthetic voices can be introduced and studied, alongside more diverse text resources which may provide deeper insight in terms of vocabulary coverage and model robustness.

Finally, this thesis seeks to contribute to the enrichment of available resources for

Romanian, especially within specialized domains such as astrology, thereby facilitating the development of various practical speech-related applications such as summarization, language identification, transcription, translation.

Bibliography

[1] Rosana Ardila, Megan Branson, Kelly Davis, Michael Henretty, Michael Kohler, Josh Meyer, Reuben Morais, Lindsay Saunders, Francis M. Tyers, and Gregor Weber. *Common Voice: A Massively-Multilingual Speech Corpus*. 2020. arXiv: 1912.06670 [cs.CL]. URL: https://arxiv.org/abs/1912.06670.

[2] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. *wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations*. 2020. arXiv: 2006.11477 [cs.CL]. URL: https://arxiv.org/abs/2006.11477.

[3] Alexis Conneau, Min Ma, Simran Khanuja, Yu Zhang, Vera Axelrod, Siddharth Dalmia, Jason Riesa, Clara Rivera, and Ankur Bapna. *FLEURS: Few-shot Learning Evaluation of Universal Representations of Speech*. 2022. arXiv: 2205.12446 [cs.CL]. URL: https://arxiv.org/abs/2205.12446.

[4] yt-dlp contributors. *yt-dlp: a youtube-dl fork with additional features and fixes*. https://github.com/yt-dlp/yt-dlp.

[5] Cong-Thanh Do, Shuhei Imai, Rama Doddipatla, and Thomas Hain. *Improving Accented Speech Recognition using Data Augmentation based on Unsupervised Text-to-Speech Synthesis*. 2024. arXiv: 2407.04047 [cs.CL]. URL: https://arxiv.org/abs/2407.04047.

[6] Dawit Ketema Gete, Bedru Yimam Ahmed, Tadesse Destaw Belay, Yohannes Ayana Ejigu, Sukairaj Hafiz Imam, Alemu Belay Tessema, Mohammed Oumer Adem, Tadesse Amare Belay, Robert Geislinger, Umma Aliyu Musa, Martin Semmann, Shamsuddeen Hassan Muhammad, Henning Schreiber, and Seid Muhie Yi-

mam. *Whispering in Amharic: Fine-tuning Whisper for Low-resource Language*. 2025. arXiv: 2503.18485 [cs.CL]. URL: https://arxiv.org/abs/2503.18485.

[7] Google Cloud. *Supported voices and languages | Cloud Text-to-Speech API*. Accessed: 2025-06-16. 2024. URL: https://cloud.google.com/text-to-speech/docs/list-voices-and-types.

[8] Hugging Face. *WER - a Hugging Face Space by evaluate-metric*. Accessed June 13, 2025. URL: https://huggingface.co/spaces/evaluate-metric/wer (visited on 06/13/2025).

[9] Hugging Face. *Whisper Fine-Tuning Event*. 2022. URL: https://github.com/huggingface/community-events/tree/main/whisper-fine-tuning-event#fine-tune-whisper.

[10] John Levis and Ruslan Suvorov. "Automatic Speech Recognition". In: Nov. 2012. ISBN: 9781405194730. DOI: 10.1002/9781405198431.wbeal0066.

[11] Jérôme Louradour. *whisper-timestamped*. https://github.com/linto-ai/whisper-timestamped. 2023.

[12] Andrew Morris, Viktoria Maier, and Phil Green. "From WER and RIL to MER and WIL: improved evaluation measures for connected speech recognition." In: Jan. 2004.

[13] Aaron van den Oord, Sander Dieleman, Heiga Zen, Karen Simonyan, Oriol Vinyals, Alex Graves, Nal Kalchbrenner, Andrew Senior, and Koray Kavukcuoglu. *WaveNet: A Generative Model for Raw Audio*. 2016. arXiv: 1609.03499 [cs.SD]. URL: https://arxiv.org/abs/1609.03499.

[14] OpenAI et al. *GPT-4 Technical Report*. 2024. arXiv: 2303.08774 [cs.CL]. URL: https://arxiv.org/abs/2303.08774.

[15] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. *Robust Speech Recognition via Large-Scale Weak Supervision*.

2022. arXiv: 2212.04356 [eess.AS]. URL: https://arxiv.org/abs/2212.04356.

[16] Leonard Richardson. “Beautiful soup documentation”. In: *April* (2007).

[17] Neti Sandu. *ProTv Horoscop*. https://stirileprotv.ro/show-buzz/horoscop/horoscop-13-iunie-2025-cu-neti-sandu-zodia-care-isi-va-schimba-locul-de-munca-in-urmatoarea-perioada.html.

[18] Silero Team. *Silero Models: pre-trained enterprise-grade STT / TTS models and benchmarks*. https://github.com/snakers4/silero-models. 2021.

[19] Silero Team. *Silero VAD: pre-trained enterprise-grade Voice Activity Detector (VAD), Number Detector and Language Classifier*. 2024.

[20] *Text to Speech - Azure Cognitive Services*. Microsoft. URL: https://learn.microsoft.com/en-us/azure/cognitive-services/speech-service/text-to-speech (visited on 06/13/2025).

[21] Vincenzo Timmel, Claudio Paonessa, Reza Kakooee, Manfred Vogel, and Daniel Perruchoud. *Fine-tuning Whisper on Low-Resource Languages for Real-World Applications*. 2025. arXiv: 2412.15726 [cs.CL]. URL: https://arxiv.org/abs/2412.15726.

[22] Wikipedia contributors. *Word error rate — Wikipedia, The Free Encyclopedia*. [Online; accessed 12-June-2025]. 2025. URL: https://en.wikipedia.org/w/index.php?title=Word_error_rate&oldid=1281062858.

[23] Xabier de Zuazo, Eva Navas, Ibon Saratxaga, and Inma Hernáez Rioja. *Whisper-LM: Improving ASR Models with Language Models for Low-Resource Languages*. 2025. arXiv: 2503.23542 [cs.CL]. URL: https://arxiv.org/abs/2503.23542.

Appendix A

Prediction examples

Table A.1: **Prediction example - common types of errors.** *Prediction example extracted from the RealVoiceHoroscope test set that displays diverse types of errors, numeric-related as well.*

Source	**Normalized prediction**	**WER (%)**
Model fine-tuned on (400-1-Standard-B)	mult prea profunde și mult prea importante pe care s ar putea să nu le puteți gestiona mulți dintre voi iar dacă este așa și nu vă gestionați emoțiile atunci să știți că nu e vina nimănui este lecția vastră de gestionare emoțională și de a nu mai căuta vinovați că vorba aceea luna plină este în scorpion apoi pe trei spreszece mai este o zi foarte importantă în care venus vă ajută foarte mult	4.05
Model fine-tuned on (400-1-EmilNeural)	mult prea profund de și mult prea importante pe care s ar putea să nu le puteți gestiona mult dintre voi iar dacă este așa și nu vă gestionați emoțiile atunci să știți că nu e vina nimă nu i este lecția văstă de gestionare emoțională și de anumă ai căuta vinovați că vorba ce a lună plină este în scorpion apoi pe treiți prăzece mai este o zi foarte importantă în care venus vă ajută foarte mult	20.27
Model fine-tuned on (800-1-EmilNeural)	mult prea profunde și mult prea importante pe care s ar putea să nu le puteți gestiona mulți dintre voi iar dacă este așa și nu vă gestionați emoțiile atunci să știți că nu e vina nu i mănui este lecția vastră de gestionare emoțională și de a nu mai căuta vinovați că vorba aceea luna plină este în scorpion apoi pe treisprezece mai este o zi foarte importantă în care venus vă ajută foarte mult	5.40
Model fine-tuned on (1600-1-EmilNeural)	mult prea profunde și mult prea importante pe care s ar putea să nu le puteți gestiona mulți dintre voi iar dacă este așa și nu vă gestionați emoțiile atunci să știți că nu e vina nimănui este lecția vastră de gestionare emoțională și de a nu mai căuta vinovați că vorba aceea luna plină este în scorpion apoi pe treisprezece mai este o zi foarte importantă în care venus vă ajută foarte mult	1.35

Table A.2: **Prediction example - specific terms correctly transcribed.** Prediction example extracted from the RealVoiceHoroscope test set that displays correctly identified astrology-related terms by the models fine-tuned on augmented corpora.

Source	**Normalized prediction**	**WER (%)**
Ground-truth	și practic jupiter în rac poate să aibă efectul ăsta de a ne face ca toți să ne autoizolăm în propriile noastre cercuri limite bun eu recomand pe jupiter în rac să îți cultivi partea asta de viață privată de a ți construi să zicem acele ancore în viața	0
Model fine-tuned on initial training set (no augmentation)	și practic jubită în rac poate să e b efectul ăsta de an face ca ați dut să ne autoizolăm în proprile noastre cercuri limite bune eu recomand pe jubită în rac să îți cultiv partea asta de viețã privată de ați construi să zicem acele ancoare în viața	30.61
Model fine-tuned on (2000-1-EmilNeural)	și practic jupiter în rac poate să i defect o stadia neface ca toți să ne autoizolăm în proprile noastre ceri cu limite bune eu recomand pe jupiter în rac să îți cultivi partea asta de vieță privată de ați construi să zicem acele ancori în viața	30.61